

LESSON 7-2 • SESSION 1 • PRACTICE • FORM B

Rational Numbers on a Number Line

Learning target – I can plot a rational number and its opposite on a number line. · 6.NS.C.6c

REMEMBER

- A **rational number** can be written as a fraction of two integers: $3/4$, $-2.5 = -5/2$, $7 = 7/1$.
- To plot one, mark the integers, then split each gap into **equal parts** (halves, fourths, tenths).
- Negative numbers sit **left of 0**, or **below 0** on a vertical number line.

WORKED EXAMPLEPlot 1.5 and -3.5 on a number line.**Step 1** Mark the integers from -4 to 4 .**Step 2** Split each gap into halves.**Step 3** 1.5 is halfway between 1 and 2. -3.5 is halfway between -4 and -3 .**Answer** 1.5 is right of 0; -3.5 is left of 0**WORDS** rational number (*número racional*) · integer (*número entero*) · number line (*recta numérica*)**PRACTICE**

show your work · label units

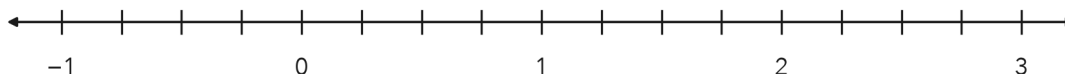
1. Name the two integers each number is between.

a. $-7/2$ _____

b. 0.4 _____

c. $-9/4$ _____d. -3.7 _____

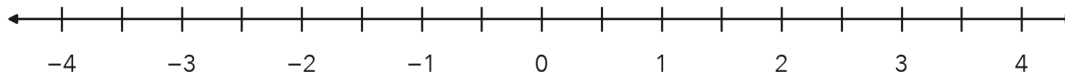
- 2.
- Garden Log.**
- Ms. Patel's class measures from the soil surface (0): sunflower 2.5 m up, carrot root 0.5 m down, bean sprout 1.25 m up, tulip bulb 0.3 m down.



a. Write a rational number for each: sunflower ____ carrot ____ bean ____ tulip ____

b. Plot and label all four on the number line.

- 3.
- Lab Thermometer.**
- A lab thermometer read
- -3.5
- degrees at 6 a.m. and 3.5 degrees at noon. Plot both readings. How far is each reading from 0?

Hint Count the halves from 0 to each point.**Answer** _____ degrees

- 4.
- Mixing Bowl.**
- A cook added
- $7/4$
- cups of broth to a bowl, then removed
- $7/4$
- cups. Write the removal as a rational number in decimal form.

Hint Removing makes the number negative. $7/4 = 1\frac{3}{4}$.

Show your work.

Answer _____ cups

- 5.
- Rational Claim.**
- Jalen says
- -4
- is not a rational number because it is not written as a fraction. Write
- -4
- as a fraction with a denominator of 1. Is Jalen right?

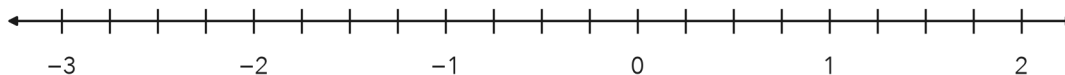
Hint Any integer can be divided by 1.

Show your work.

Fraction _____

6. **Three Marks.** A number strip shows the marks $-9/4$, 1.6 and -0.35 . Plot all three. Which mark is closest to 0?

Hint Write $-9/4$ as a decimal first: -2.25 .



Closest to 0 _____

7. **Pool Depths.** A pool's surface is 0. A float sits 0.6 m above the surface and a drain sits at -2.1 m. How many meters farther from the surface is the drain than the float?

Hint Find each distance from 0, then subtract.

Show your work.

Answer _____ m

8. **Ledge and Cave.** A rock ledge is 8.5 meters above the sea surface. A cave floor sits at the opposite value.

- a. Write the number for the cave floor. _____ m
- b. Between which two integers is the cave floor? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Jalen's Work.** To plot -3.6 , Jalen counted six tenths to the RIGHT of -3 and put his point between -3 and -2 . Explain his mistake. Then name the two integers -3.6 is between.

TEST PRACTICE

choose the best answer

10. On a number line, -1.5 sits between –
- A** -1 and 0 **B** 1 and 2 **C** -2 and -1 **D** -1.5 and -1
11. Which number is farthest right on a number line?
- A** -0.5 **B** -3 **C** 0.25 **D** -0.75

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

0.35 J	3.5 P	1.75 Q	-1.75 O	1.5 T	8.5 X	-8.5 S	-4/1 I	-0.35 N	-3.5 Z
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#3	#4	#5	#6	#7	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____